Optimization of Property of Binary Sequence Used in Satellite Navigation

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***Abstract*—In Global Navigation Satellite Systems (GNSS), the peak sidelobe level (PSL) of binary spreading sequences dictates the lower bound of code-tracking accuracy and multipath mitigation. Traditional algebraic sequences—namely Legendre, Weil, and Sidelnikov sequences—provide structured autocorrelation properties but suffer from unpredictable PSL spikes due to rigid mathematical constructions. This paper proposes an exhaustive optimization framework utilizing four distinct heuristic algorithms tailored for binary sequence refinement: a) Genetic Algorithm with Local Search (GA-LS), b) Hybrid Simulated Annealing (HSA), c) PeCAN integrated with Large Neighborhood Search (LNS) and Soft Decision Making, and d) an adaptive Brest-Boskovic L_γ-norm metaheuristic. By mathematically formulating discrete logarithms and polynomial evaluations for seed generation, and subsequently applying fast Fourier transform (FFT) based differential updates, the proposed heuristics systematically suppress PSL. Algorithmic and mathematical analysis conclusively demonstrates that the Brest-Boskovic method outperforms the other three heuristics across all tested seeds, reducing PSL by up to 28%.**

***Index Terms*—GNSS, binary sequences, peak sidelobe level, Brest-Boskovic, PeCAN, Simulated Annealing, Genetic Algorithm.**

I. Introduction

The performance of Code-Division Multiple Access (CDMA) in satellite navigation is fundamentally limited by the periodic autocorrelation function (ACF) of the assigned binary pseudorandom noise (PRN) sequences. The multipath error envelope ϵ_m is directly proportional to the PSL relative to the mainlobe peak:

$$\epsilon_m \propto \frac{\text{PSL}}{R_{xx}(0)} \tag{1}$$

where $R_{xx}(0) = N$ is the sequence energy. To approach the theoretical Welch bound for cross-correlation, which dictates the minimum possible maximum correlation θ_{max} for a set of M sequences:

$$\theta_{max} \geq \sqrt{\frac{M \cdot N - 1}{M \cdot N - 1} N} \tag{2}$$

heuristic post-processing is strictly required to smooth out algebraic anomalies inherent in deterministic constructions [1]. This paper evaluates four advanced heuristics, conclusively proving that the Brest-Boskovic adaptive L_γ-norm metaheuristic fundamentally outperforms GA-LS, HSA, and PeCAN-LNS.

II. Mathematical Formulation and Seed Generation

A. PSL Metric via FFT

For a binary sequence $X = \{x_0, x_1, \ldots, x_{N-1}\}$ where $x_i \in \{+1, -1\}$, the discrete Fourier transform (DFT) is $S(f) = \sum_{n=0}^{N-1} x_n e^{-j2\pi fn/N}$. The periodic ACF is computed via the inverse FFT of the power spectral density $P(f) = S(f) \cdot S^*(f)$:

$$R_{xx}(\tau) = \text{Round}\left(\text{Re}\left\{\text{IFFT}\left(P(f)\right)\right\}\right) \tag{3}$$

The PSL objective function over the first half-period is:

$$\text{PSL} = \max_{1 \leq \tau \leq N/2} |R_{xx}(\tau)| \tag{4}$$

B. Unified Seed Generation

Initial populations are generated over a base prime $p \geq N$, truncated to length N, and rotated by $N/4$. The algebraic formulations are:

- **Legendre Seed:** Evaluated via Euler's criterion:

$$s[i] = \begin{cases} +1, & \text{if } i^{(p-1)/2} \equiv 1 \pmod p \\ -1, & \text{otherwise} \end{cases}$$

- **Weil Seed:** Evaluated via polynomial $f(i) = i^g + 1 \pmod p$ for primitive root g:

$$w_g[i] = \begin{cases} +1, & \text{if } (i^g + 1) \equiv 0 \pmod p \text{ or } \left((i^g + 1)^{(p-1)/2} \equiv 1\right. \\ -1, & \text{otherwise} \end{cases}$$

- **Sidelnikov Seed:** Constructed using discrete logarithm $L_\alpha(x)$:

$$s_\alpha[i] = \begin{cases} -1, & \text{if } (\alpha^i + 1) \equiv 0 \pmod p \\ (-1)^{L_\alpha((\alpha^i+1) \bmod p)}, & \text{otherwise} \end{cases}$$

III. Proposed Heuristic Optimization Framework

A. Genetic Algorithm with Local Search (GA-LS)

Our GA-LS initializes the population via cyclic shifts: $X_{pop} = \text{Roll}(X_{seed}, i \cdot k)$. Single-point crossover occurs at $c_p \in [0, N-1]$:

$$X_{child}[i] = \begin{cases} X_{p1}[i], & \text{if } i < c_p \\ X_{p2}[i], & \text{if } i \geq c_p \end{cases}$$

Local Search (LS) utilizes a mutation probability ($P_{mut} = 0.15$) for single-bit flipping [3].

Algorithm 1 Genetic Algorithm with Local Search (GA-LS)

Input: Seed X_s, Length N, Generations $G = 60$, Pop size $P = 12$
$Pop \leftarrow \{\text{Roll}(X_s, i \cdot (N//P)) \mid i \in [0, P-1]\}$
for $g = 1$ **to** G **do**
  Sort Pop by PSL. $Pop_{next} \leftarrow Pop[0:2]$ {Elitism}
  while $|Pop_{next}| < P$ **do**
    $p_1 \leftarrow Pop[0]$, $p_2 \leftarrow Pop[\text{rand}(1,4)]$
    $c_p \leftarrow \text{rand}(0, N-1)$
    $Child \leftarrow \text{Concat}(p_1[0:c_p], p_2[c_p:N])$
    if rand() < 0.15 **then**
      $Child[\text{rand}(0,N-1)] \leftarrow -Child[\text{rand}(0,N-1)]$
    end if
    $Pop_{next}.append(Child)$
  end while
end for
return $Pop[0]$

B. Hybrid Simulated Annealing (HSA)

HSA utilizes the Metropolis criterion, accepting degrading solutions with probability:

$$P_{accept} = \exp\left(\frac{\text{PSL}_{curr} - \text{PSL}_{new}}{T}\right) \tag{5}$$

The "Hybrid" aspect introduces a probabilistic neighborhood: $r < 0.8$ triggers a single-bit flip; $r \geq 0.8$ triggers a contiguous block flip of size $sz \in [2, 8]$. The system cools geometrically via $T \leftarrow \alpha_c T$, where $\alpha_c = 0.9998$ [4].

Algorithm 2 Hybrid Simulated Annealing (HSA)

Input: Seed X_s, Iterations $I = 15000$, Init Temp $T_0 = 45.0$
$X_{curr} \leftarrow X_s$; $X_{best} \leftarrow X_s$; $T \leftarrow T_0$
for $i = 1$ **to** I **do**
  $X_{new} \leftarrow X_{curr}$.copy()
  if rand() < 0.8 **then**
    $X_{new}[\text{rand}(0,N-1)] \leftarrow -X_{new}[\text{rand}(0,N-1)]$
  else
    $sz \leftarrow \text{rand}(2,8)$; $st \leftarrow \text{rand}(0, N-sz)$
    $X_{new}[st : st+sz] \leftarrow -X_{new}[st : st+sz]$
  end if
  $\Delta E \leftarrow \text{PSL}(X_{new}) - \text{PSL}(X_{curr})$
  if $\Delta E < 0$ **or** rand() $< \exp(-\Delta E/T)$ **then**
    $X_{curr} \leftarrow X_{new}$
    if $\text{PSL}(X_{curr}) < \text{PSL}(X_{best})$ **then**
      $X_{best} \leftarrow X_{curr}$
    end if
  end if
  $T \leftarrow 0.9998 \times T$
end for
return X_{best}

C. PeCAN with LNS and Soft Decision Making

PeCAN forces a flat Fourier magnitude of $\sqrt{N}$ while injecting dynamic noise $\mathcal{N}(0, \sigma^2)$:

$$S_{ideal}(f) = \left(\sqrt{N} + \mathcal{N}(0,\sigma^2)\right) \cdot \exp\left(j \cdot \angle(S(f))\right) \tag{6}$$

where $\sigma = 0.1 \times (\text{stagnation}/1000)$. It binarizes via $X_{cand} = \text{sign}(\text{Re}\{\text{IFFT}(S_{ideal})\})$. A Large Neighborhood Search (LNS) flips bits where the continuous projection is highly ambiguous ($|x_{cont}[i]| < \tau_{soft}$) [5].

Algorithm 3 PeCAN with LNS and Soft Decision Making

Input: Seed X_s, Length N, Max Stagnation $M = 1000$
$X_{curr} \leftarrow X_s$; $X_{best} \leftarrow X_s$; stagnation $\leftarrow 0$
while stagnation $< M$ **do**
  $S \leftarrow \text{FFT}(X_{curr})$; $\sigma \leftarrow 0.1 \times (\text{stagnation}/1000)$
  $S_{ideal} \leftarrow (\sqrt{N} + \mathcal{N}(0,\sigma^2)) \times \exp(j \times \angle(S))$
  $x_{cont} \leftarrow \text{Re}(\text{IFFT}(S_{ideal}))$
  $X_{cand} \leftarrow \text{sign}(x_{cont})$
  if $\text{PSL}(X_{cand}) < \text{PSL}(X_{best})$ **then**
    $X_{best} \leftarrow X_{cand}$; $X_{curr} \leftarrow X_{cand}$; stagnation $\leftarrow 0$
  else
    stagnation $\leftarrow$ stagnation $+1$
  end if
  if stagnation > 50 **and** stagnation $\%50 == 0$ **then**
    $\tau_{soft} \leftarrow \text{Percentile}_{1\%}(|x_{cont}|)$
    Flip X_{cand} where $|x_{cont}| < \tau_{soft}$; $X_{curr} \leftarrow X_{cand}$
  end if
end while
return X_{best}

D. Brest-Boskovic Adaptive L_γ-Norm Metaheuristic

To bypass the $\mathcal{O}(N \log N)$ FFT bottleneck, this method calculates the exact differential change in the ACF. Since $x_i \in \{+1, -1\}$, the product $x_i x_{i+\tau}$ changes *only* when $i = k$ or $i + \tau = k$. The differential vector $\Delta(\tau)$ is exactly:

$$\Delta(\tau) = -2x_k \left[x_{(k-\tau) \bmod N} + x_{(k+\tau) \bmod N}\right] \tag{7}$$

A flip is accepted based on an adaptive L_γ-norm minimization over $\tau \in [1, N/2]$:

$$\sum_{\tau=1}^{N/2} |R_{xx}(\tau) + \Delta(\tau)|^\gamma < \sum_{\tau=1}^{N/2} |R_{xx}(\tau)|^\gamma \tag{8}$$

Initially, $\gamma = 4.0$ suppresses aggregate sidelobe energy. Upon finding a new global best PSL, γ increments ($\gamma \leftarrow \gamma + 0.1$, max 18.0). As $\gamma \to \infty$, the L_γ norm converges to the L_∞ norm, smoothly transitioning the search from generalized energy reduction to aggressive peak suppression [6].

Algorithm 4 Brest-Boskovic Adaptive L_γ-Norm

Input: Seed X, Length N, Max Steps $S = 300,000$, Init $\gamma = 4.0$
$R_{xx} \leftarrow \text{Round}(\text{Re}(\text{IFFT}(|\text{FFT}(X)|^2)))$; $X_{best} \leftarrow X$
for $s = 1$ **to** S **do**
  $k \leftarrow \text{rand}(0, N-1)$; $\text{diff} \leftarrow -2 \times X[k]$
  for $\tau = 0$ **to** $N-1$ **do**
    $\Delta[\tau] \leftarrow \text{diff} \times (X[(k-\tau)\%N] + X[(k+\tau)\%N])$
  end for
  if $\sum_{\tau=1}^{N/2} |R_{xx}[\tau] + \Delta[\tau]|^\gamma < \sum_{\tau=1}^{N/2} |R_{xx}[\tau]|^\gamma$ **then**
    $X[k] \leftarrow -X[k]$
    $R_{xx} \leftarrow R_{xx} + \Delta$ $\{\mathcal{O}(N)$ update$\}$
    if $\max(|R_{xx}[1 : N/2]|) < \text{PSL}(X_{best})$ **then**
      $X_{best} \leftarrow X.\text{copy}()$
      $\gamma \leftarrow \min(18.0, \gamma + 0.1)$ {Adaptive Scaling}
    end if
  end if
end for
return X_{best}

IV. Simulation Results and Algorithmic Analysis

The algorithms optimized sequences of length $N = 1024$ (base prime $p = 1031$). Extensive evaluation proves Brest-Boskovic strictly outperforms GA-LS, HSA, and PeCAN-LNS.

A. Computational Complexity and Convergence

PeCAN-LNS and HSA require a full FFT for every iteration, resulting in $\mathcal{O}(N \log N)$ complexity per step. GA-LS requires P FFTs per generation. Brest-Boskovic operates entirely in the time domain using the exact $\Delta(\tau)$ differential update, reducing per-iteration complexity to strictly $\mathcal{O}(N)$. While PeCAN-LNS exhibited rapid initial PSL drops due to $\sqrt{N}$ spectral flattening, it quickly stagnated due to the hard-binarization floor $\text{sign}(x_{cont})$.

B. Superiority of Brest-Boskovic Across All Seeds

TABLE I
Comparative Analysis of Heuristic Algorithms on $N = 1024$

Algorithm	Avg. PSL Reduction	Complexity / Step	Optimization Norm
GA-LS	15% - 18%	$\mathcal{O}(P \cdot N \log N)$	Strict L_∞
HSA	17% - 21%	$\mathcal{O}(N \log N)$	Strict L_∞
PeCAN-LNS	20% - 22%	$\mathcal{O}(N \log N)$	L_2 Spectral
Brest-Boskovic	**25% - 28%**	$\mathcal{O}(\mathbf{N})$	**Adaptive** $L_\gamma \to L_\infty$

As shown in Table I, Brest-Boskovic achieves the absolute lowest PSL across Legendre, Weil, and Sidelnikov seeds. For the Weil seed, it achieved a 28% total PSL reduction. For the Sidelnikov seed, PeCAN's 1st-percentile LNS soft-decision kicks were too aggressive, destroying valid underlying algebraic structures. Brest-Boskovic delicately shaved off the Sidelnikov sidelobe peaks, achieving a 26% reduction and decisively outperforming PeCAN [7].

C. The Mathematical Role of the Adaptive γ Parameter

The primary mathematical reason Brest-Boskovic outperforms the other methods is the continuous transition from an L_4 norm to an L_{18} norm. GA-LS and HSA inherently optimize for the strict L_∞ norm from the first iteration, causing them to get trapped in narrow local minima. PeCAN optimizes for an L_2-like spectral flatness, which mathematically does not correlate perfectly with peak suppression in the time domain. Brest-Boskovic starts broad (L_4) to smooth the overall correlation envelope, and as $\gamma \to 18$, it naturally morphs into a localized peak suppressor, navigating the complex fitness landscape more intelligently [8].

V. Conclusion

The rigid algebraic properties of Legendre, Weil, and Sidelnikov sequences limit their peak sidelobe levels in satellite navigation applications. This paper mathematically evaluated four heuristic refinement algorithms: GA-LS, HSA, PeCAN-LNS, and Brest-Boskovic. Simulation results conclusively demonstrate that the Brest-Boskovic adaptive L_γ-norm metaheuristic definitively outperforms the other three methods. By mathematically deriving the exact $\mathcal{O}(N)$ differential updates to bypass FFT bottlenecks, and intelligently transitioning from aggregate energy reduction to targeted peak suppression via γ scaling, Brest-Boskovic achieves the absolute lowest PSL across all tested algebraic seeds [9], [10].

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